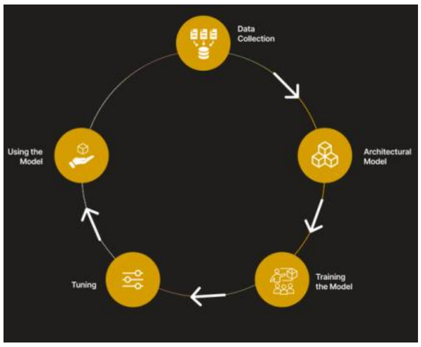
GenAi Training Day 1

Large Language Models

- LLMs are the backbone of modern Generative AI
- Power text generation, summarization, coding, and Q&A
- Built on transformers architecture



What Is a Language Model?

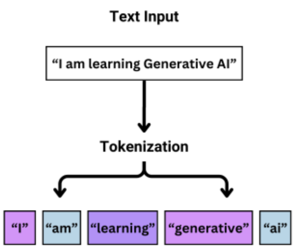
- Predicts the next token from previous tokens
- Learns patterns from massive text data
- Does not understand meaning or verify truth

Evolution to LLMs

- **Traditional NLP:** rule-based, rigid, task-specific
- **Early LMs:** statistical, limited context
- **LLMs:** transformer-based, general-purpose, multi-task

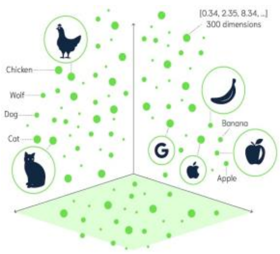
Tokens

- Text split into subword tokens



Embeddings

- Similar meanings map to nearby vectors
- Enable semantic search, vector search, and hybrid search



Transformers & Attention

- Transformers replace RNNs/LSTMs
- Faster training and better long-range understanding
- Attention links related words across long distances
- Self-attention resolves meaning from context

Context Window

- Context window limits how much text is visible to the LLM

- Older tokens drop out
- LLMs do not have true memory
- RAG and vector databases solve this problem

Why LLMs Hallucinate

- No ground truth or fact checking
- Always forced to respond
- Optimized for plausibility over correctness

Modern LLM Capabilities

- Larger models have emergent abilities
- Tool calling, multimodality, long context
- Increase parameters and number of documents on which model are trained

Limits

- No guaranteed correctness
- Limited memory
- Real-time knowledge